COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling

Lecture 16: Poisson Regression

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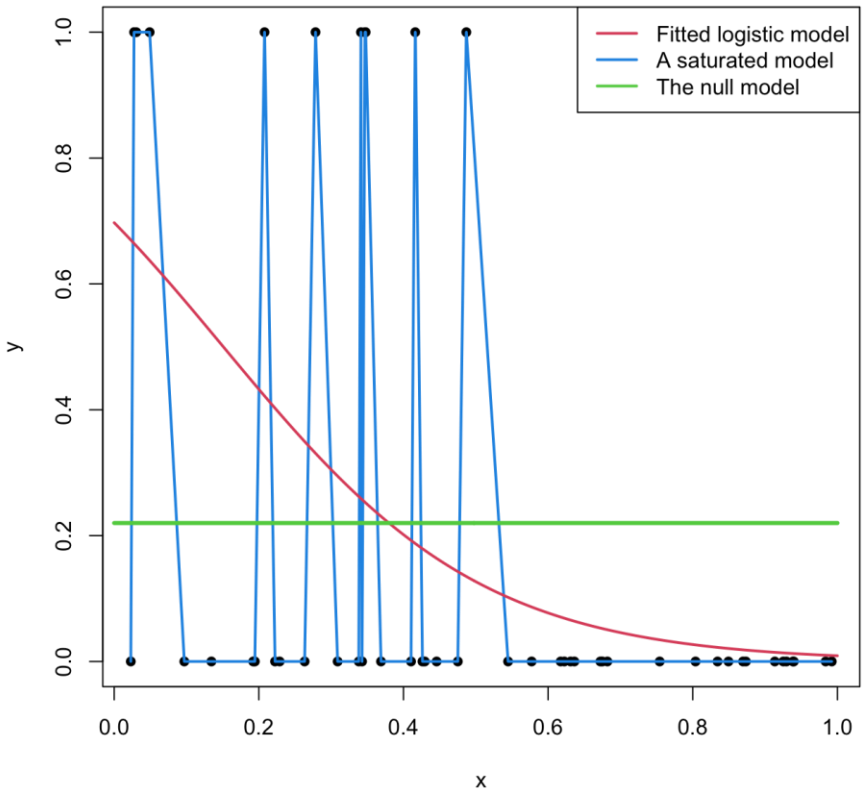
From last time: Deviance

It measures the difference of the fitted generalized linear model with respect to a perfect model for the sample.
This perfect model, known as the *saturated model*, is the model that perfectly fits the data, in the sense that the fitted responses equal the observed responses.
For a linear model, deviance is the sum of squared errors (SSE) and D0 is the total sum of squares (SST).

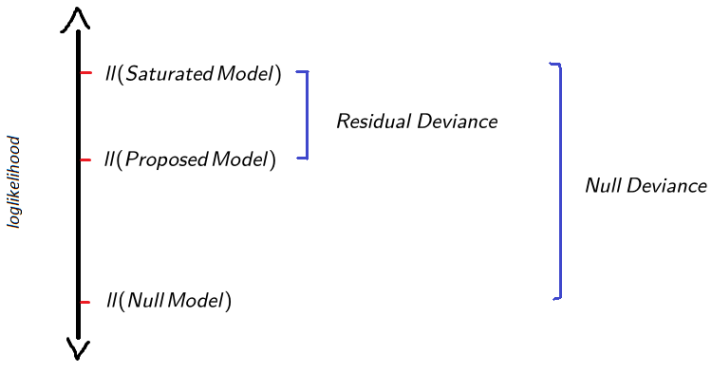


Deviance (continued)

- The deviance for a GLM m is defined as

$$D(\boldsymbol{y}, m) = -2\left[l(\widehat{\boldsymbol{\beta}}) - l_S\right]$$

- It measures the difference of the fitted generalized linear model with respect to a perfect model for the sample.
- In the linear case, D = SSE.
- In R, the "Residual Deviance" is two times the difference in the log likelihood of the saturated (perfect) model and our model.
- In R, Null Deviance = 2[LL(Saturated Model) - LL(Null Model)]
- Note, deviance is not equal to KL distance
- But $D(\boldsymbol{y}, m_2) - D(\boldsymbol{y}, m_1) = KL(\boldsymbol{y}, m_2) - KL(\boldsymbol{y}, m_1)$
- See Hastie (1987) article in The American Statistician for more details.



Residuals

- Raw residuals e_i, aka response residuals are of limited use as each has its own standard error.
- In general, when using GLM, it is better to request *Pearson residuals*, which are equal to the raw residual divided by an estimate of its standard error.
- For logistic regression these are

$$r_i = \frac{y_i - \hat{\pi}_i}{\sqrt{\hat{\pi}_i(1 - \hat{\pi}_i)}}$$

- Note that there are also *deviance* residuals d_i such that $D = \sum_{i=1}^{n} d_i^2$
- For logistic regression,

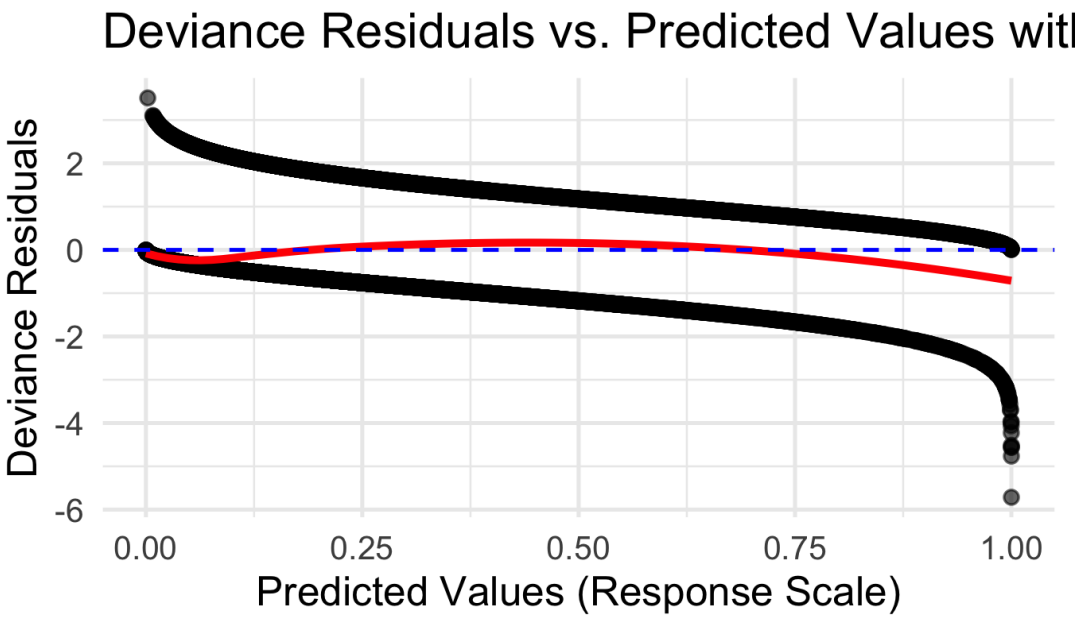
$$d_i = sign(e_i)\sqrt{-2(y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)}$$

In R use: `resid(fittedmodel, type = "dev"))`

Q: What can we do with the residuals?

Diagnostics

- Influential observations can be detected in similar way as in regular linear regression.
- Plot of observed vs. fitted frequencies is also helpful.



Introduction

Recall: Logistic regression model is applicable when the response variable Y is *dichotomous* (that is, Y takes only *two* values).

Now: Response variable Y represents *counts* (that is, takes only whole integer values).

Summary of Generalized Linear Models (GLM):

- Regular linear regression - when Y is continuous numerical.
- Logistic regression - when Y is categorical with only two outcomes.
- Poisson regression - when Y is a count.
- Multinomial or ordinal regression when Y is categorical with more than two outcomes.
- Gamma regression – when the outcome variable is positive, like amount of an insurance claim.

Poisson Data Examples

- Number of patients in an emergency room at any given time. Possible predictors: time, day of the week, season, …

- Number of awards won by a movie. Predictors include number of stars in the movie, budget, IMDB rating, …

- Number of children per family, based on income, race/ethnicity, place of residence (urban or rural), …

Assumptions

- Observations are independent (standard statistical assumption except in time series analysis and Markov chains)

- The distribution of Y (conditional on the $\boldsymbol{X}$ variables) follows a Poisson distribution:

$$Y|\boldsymbol{X} \sim Poisson\big(\lambda(\boldsymbol{X})\big)$$

Recall that $\lambda > 0$ is a parameter of the distribution equal to *both* the mean and the variance

- Because of the properties of the Poisson distribution, the Poisson regression model is known to satisfy the *equidispersion* property. If data indicate *overdispersion* (variance greater than the mean), then alternatives like the negative binomial regression might be more appropriate.

Poisson Regression Model

- Log-Linear model:

$$\log[E(Y|X)] = \log(\lambda) = b_0 + b_1X_1 + \ldots + b_pX_p$$

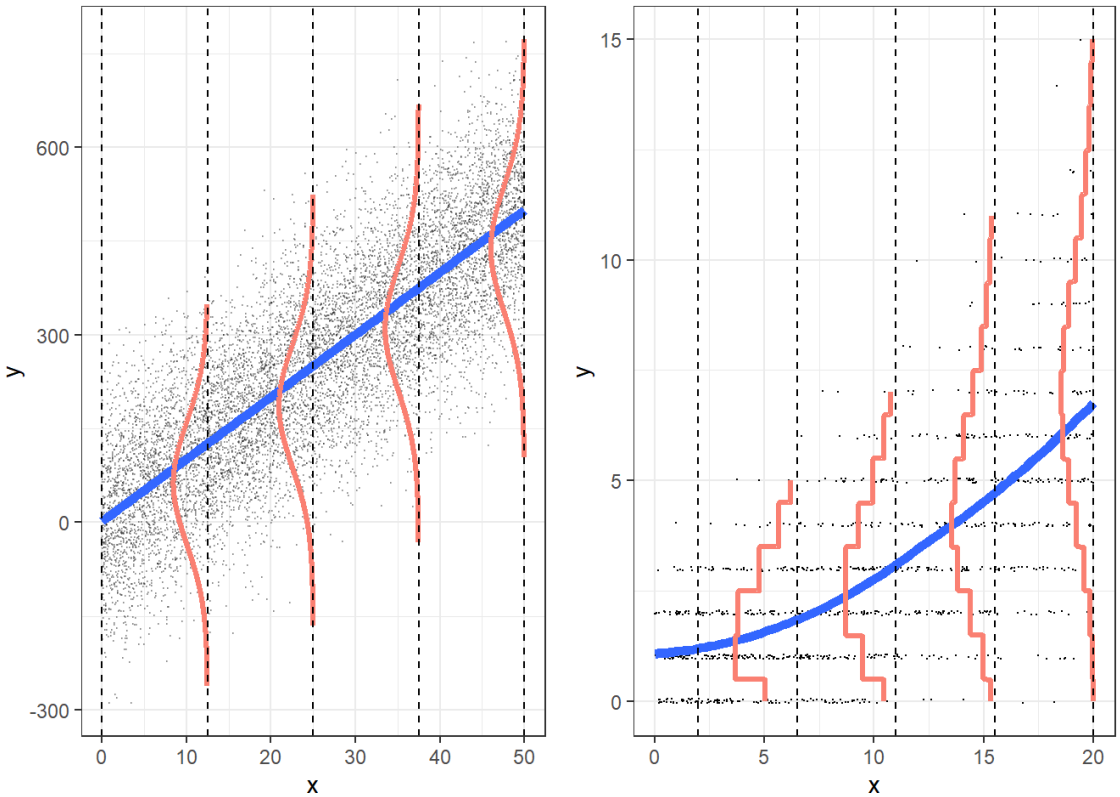
or equivalently

$$E(Y|X) = \lambda = \exp(b_0 + b_1X_1 + \ldots + b_pX_p)$$

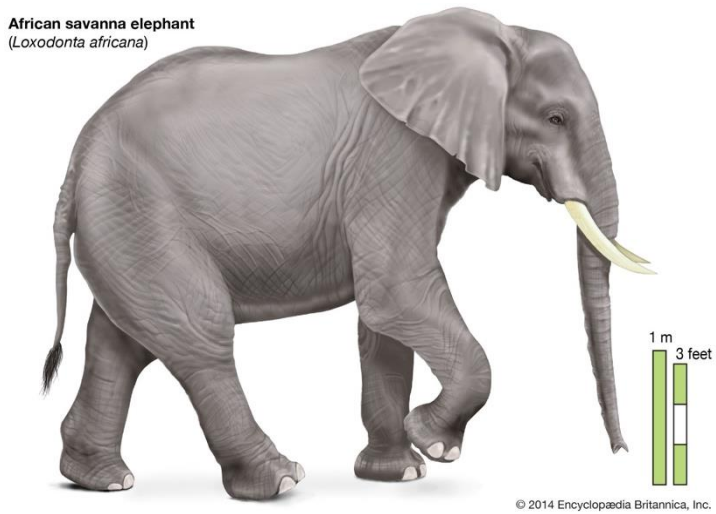
- Like the logistic regression, $\exp(b_j)$ represents the *multiplicative* effect of the jth predictor on the mean, keeping the other predictors fixed. That is, increasing X_j by one unit *multiplies* the mean of the response variable by a factor of $\exp(b_j)$.

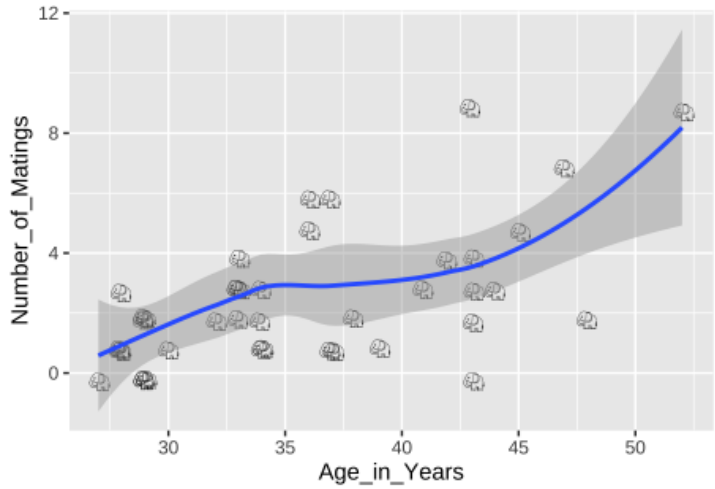
- Empirical evidence shows that with count data the effects of predictors are often multiplicative rather than additive. That is, one typically observes small effects for small counts, and large effects for large counts.

Linear regression (left) vs. Poisson regression (right).



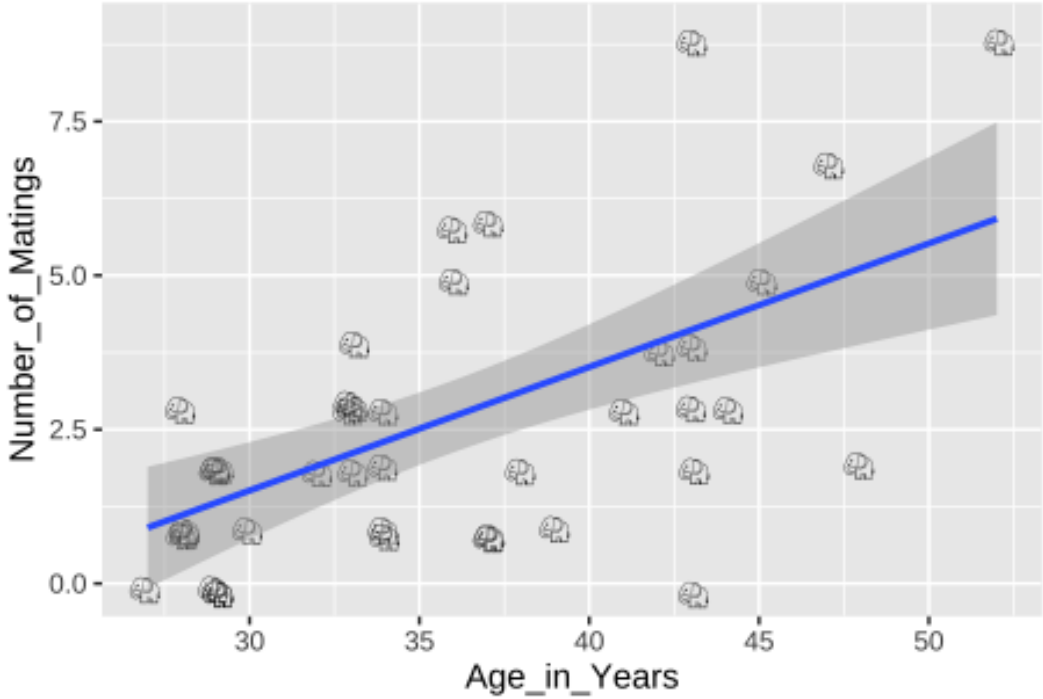
Example 1: Elephant Data





- Data from R package `gpk`.

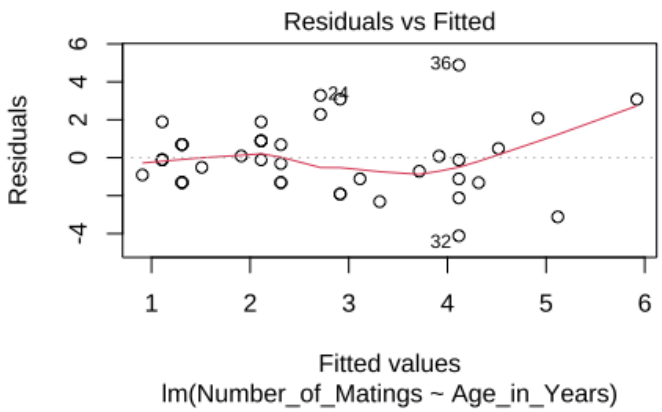
- Elephants reach maturity at about 14 years of age. But they must compete with all adult males for mating opportunity. Females are generally more receptive to larger males. Size of an elephant increases as age increases. Hence it is expected that generally the number of matings should increase with age.

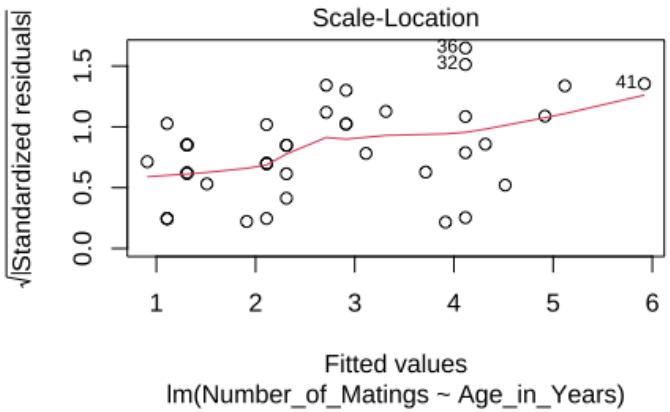


Example 1: Linear model: what can go wrong?

- Estimated equation is $\hat{Y} = -4.5 + 0.2X$
- We estimate that elephant males expect, on average, an additional 0.20 mates, with a 95% CI (0.11, 0.29), for every year they live.
- R^2 is quite poor with 34.3% explained variability in Y due to X.

Example 1: Linear model: what can go wrong?

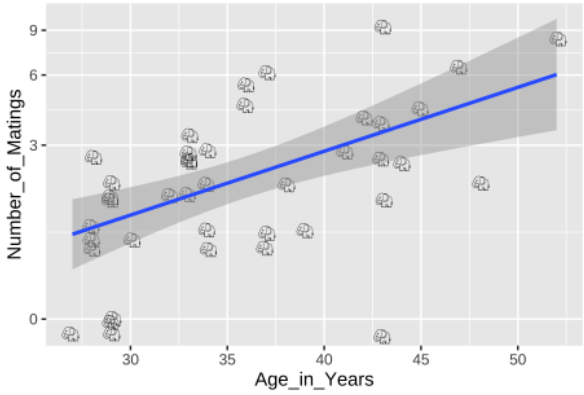




- Diagnostic plots suggest nonlinearity
- Also note that the variance in the residuals increases with the predicted value.

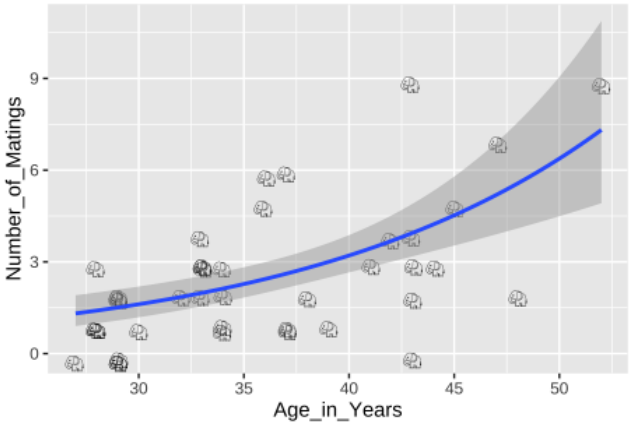
Example 1: Log-transformed data

- Estimated equation is $\hat{Y} = e^{-0.7+0.05X} - 1$
- Roughly, we estimate that male elephant expect, on average, a 5.1% increase ($e^{0.05} = 1.051$) in number of matings for every additional year they live.
- R^2 is even worse.
- Diagnostic plots show improvement.
- Why bother: we already dealt with the complexity of transforming our data for the model and then back-transforming to explain the results, so a Poisson regression might be more convenient.



Example 1: Elephant Poisson GLM

- Estimated equation is $\hat{\lambda} = e^{-1.58+0.0687X}$
- We estimate that male elephants, on average, experience a 7.1% increase ($e^{0.0687} = 1.071$) in number of matings for every additional year they live.
- Don't be tempted to compare the AIC between the different models, because they use different likelihood functions!
- Hypothesis test with p-value = 7.991e-07 shows the model fits the data better than no model (that is, significantly different than the model without any predictors).



Alternatives

Common violations of Poisson assumptions include

- Overdispersion (more variance than predicted by the Poisson)

- Zero inflation (too many zeros)

- Quasipoisson & negative binomial are usually good options for modeling overdispersion.
- In quasipoisson we assume $Var = Var(\mu) = \phi\mu$

- Mixture models can be used to model "zero inflated" data.

Example 2: Wine data (from "Applied Regression Modeling")

Wine demand at a restaurant predicted, using factors such as origin, variety, sensory descriptors (aroma, mouth feel, etc.) and price. There are 20 categorical predictors + numerical predictor price. The file contains data on the white wines only.

- *Y* = bottles sold
- *X* = price
- *D1* = indicator for having the lowest price in a menu category
- *D2* = indicator for being available by the glass
- *D3* = indicator for Oregon Chardonnay (ref: California Chardonnay)
- *D4* = indicator for Oregon Pinot Gris (ref: California Chardonnay)
- *D5* = indicator for California other (ref: California Chardonnay)
- *D6* = indicator for Northwest other (ref: California Chardonnay
- *D7* = indicator for French (ref: California Chardonnay)

Example: wine data

- *D*8 = indicator for body
- *D9* = indicator for finish
- *D*10 = indicator for oak
- *D*11 = indicator for rich
- *D12* = indicator for spice
- *D*13 = indicator for buttery
- *D*14 = indicator for creamy
- *D*15 = indicator for dry
- *D16* = indicator for honey
- *D*17 = indicator for melon
- *D*18 = indicator for citrus
- *D*19 = indicator for other tree fruit
- *D*20 = indicator for tropical fruit

Significance of the predictors

Notice there are some very highly insignificant predictors (p-values highlighted in red).

We will remove $D9$, $D11$, $D12$, $D13$, $D16$, $D17$, $D19$, and $D20$, and re-run the model to see if it fits better.

Now all predictors are significant!

There is a formal test for removing a group of predictors, like the partial F-test in linear regression.

Chi-square Deviance test

H_0: coefficients of all removed predictors are = 0

Reduced model:

Residual Deviance = 495.23

Complete model:

Residual Deviance = 492.17

Recall, the residual deviance measures how well the model fits the data (that is, compared to the saturated model).

In the reduced model deviance is 3.05 higher (worse) than complete model.

However, this is not a significant difference based on Chi-square with 8 df (= number of removed predictors), as χ^2 critical value = 15.51 (from a table).

Alternatively: compute p-value = 0.9312 (from R)

Interpretation of the Poisson Regression parameters

- Coefficient for X (the price) = -0.084:

for each extra \$1 increase in price, log[E(Y)] decreases by 0.084 units, keeping other predictors fixed. That is, expected quantity sold will multiply by exp(-0.084) = 0.92. This means there is 8% *decrease* in demand for every \$1 increase in price (as per basic economics theory).

- Coefficient for $D1$ (lowest price in its category), that is -0.912, with exp(-0.912) = 0.4017119:

there is 59.8% decrease in demand if wine is the cheapest in its category and all other predictors are the same.

- Which variety is the most popular?

$D3$-$D7$ are all negative, so when any of $D3$-$D7$ are equal to 1, demand decreases, meaning reference region California Chardonnay has the highest demand.

- What is the effect on demand if wine is available by the glass?

It increases demand as $D2$ has coefficient 1.755

Is the equidispersion assumption satisfied?

- Sample mean of Y is 1.38, while sample variance is 7.55, so it looks like assumption is violated. However, we need to adjust for the predictors.

After adjusting for the predictor variables overdispersion drops to 1.148339 (not in base R output, but there is a package for overdispersion test). This still indicates of a problem – should be equal to 1!

An alternative estimate is $\frac{\sum_{i=1}^{n} r_i^2}{n-p-1}$, where r_i are the Pearson residuals (see next slide).

- How can we fix this? Try to fit a negative binomial model (see Exercise below).

Exercise: Poisson regression with crab data

- Open the data from last time.
- Fit a model to predict the *number* of satellites based only on one predictor: shell width
- Report the regression equation

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.30476    0.54224  -6.095  1.1e-09 ***
width        0.16405    0.01997   8.216  < 2e-16 ***
```

- Log(E(Y)) = ?
- Interpret the value of the slope

Exercise (continued)

- What is the predicted *average* number of satellites for female crabs of width 26.3 cm?

- Compute also the predicted number of satellites for 27.3 cm and cross-check that the ratio of this and previous answer equals $\exp(b_1)$.

Poisson regression: practice with crab data

- Fit a negative binomial model and report the regression equation.

R code:

```
library(MASS)
summary(m1 <- glm.nb(satell ~ width, data = crabs))
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.05251    1.17143  -3.459 0.000541 ***
width        0.19207    0.04406   4.360  1.3e-05 ***
```

- Log(E(Y)) = ?

More Complicated GLMs

- We studied Gaussian, logistic and Poisson GLMs.
- Many datasets include dependent variables whose distributions cannot be represented by the normal, binomial or Poisson distributions.
- For example, distributions of common survey questions such as “How much do you like this class? (1–7)”.
- Sometimes numerical variables, like age and years of education are binned into ordered categories, the number of which may vary from survey to survey.
- In such situations, sometimes the research interest lies not in the scale of each individual variable, but rather in the associations between the variables.
- Even if the variables are not artificially binned, it is natural to think of many ordinal, non-numeric variables as arising from some underlying latent numeric process.
- For example, the severity of a disease might be described “low”, “moderate” or “high”, although we imagine a patient’s actual condition lies within a continuum.
- There are specialized, complex distribution for ordered categories, ranks or nominal data.

Ordinal Regression

Ordinal logistic regression is a statistical analysis method that can be used to model the relationship between an ordinal response variable and one or more explanatory variables.

The model is:

$$\log \frac{P(Y \leq j)}{1 - P(Y \leq j)} = \beta_{j0} - \beta_1 x_1 - \beta_2 x_2 - \cdots - \beta_p x_p, j = 1, \ldots, J - 1$$

The model incorporates a negative sign so that there is a direct correspondence between the slope and the ranking. Thus, a positive coefficient indicates that as the value of the explanatory variable increases, the likelihood of a higher-ranking increases.

Ordinal Regression Example

Suppose we are interested in describing the relationship between the educational attainment and number of children of individuals in a population. Additionally, we might suspect that an individual's educational attainment may be influenced by their parent's education level. The 1994 General Social Survey provides data on variables DEG, CHILD and PDEG for a sample of individuals in the United States, where DEG_i indicates the highest degree obtained by individual i, CHILD_i is their number of children and PDEG_i is the binary indicator of whether either parent of i obtained a college degree.

GLM: Real Data Example

Data:

Smoking	Status	Age 18-24	25-34	35-44	45-54	55-64	65-74	75+
NonSmoker	Alive	61	152	114	66	81	28	0
	Dead	1	5	7	12	40	101	64
Smoker	Alive	53	121	95	103	64	7	0
	Dead	2	3	14	27	51	29	13

Smoking and survival for women in Whikcham near Newcastle upon Tyne in northern England. (Appleton et. al. 1996)
Data collected in 1972-1974 and then follow up 20 years later.

Research question: *How does smoking affect survival rate?*